

Transparent Loan Approval Prediction: A Machine Learning Approach with SHAP-based Interpretability

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Abstract—Accurate loan approval prediction has become crucial in promoting equitable lending practices and ensuring responsible financial decision-making. Machine learning (ML) models are widely utilized in classifying loan approvals due to their strong predictive accuracy. However, the predictions made by these models are often difficult to explain and trust. This study investigates the CatBoost model using interpretable machine learning methods, specifically focusing on Shapley Additive Explanations (SHAP), to enhance the model's interpretability, transparency, and understanding of factors influencing loan approval decisions. We implemented the Catboost Model and compared its predictive performance with eight ML models, including XGBoost, LightGBM, Gradient Boosting, Random Forest, Extra Trees, Decision Tree, AdaBoost, and BaggingClassifier for predicting loan approval status. The Catboost model demonstrates impressive accuracy, achieving a remarkable accuracy of 99.85%, outperforming other classifiers such as XGBoost at 99.38% and Bagging-Classifer at 99.25%. The outputs of the most accurate model, CatBoost, were explained using model-agnostic explanation techniques called Shapley additive explanations (SHAP). The SHAP method revealed that Risk Score, Debt-to-Income Ratio, and Total Debt-to-Income Ratio are the three key features influencing loan approval. Additionally, it demonstrated how each feature impacts the predictions made by the CatBoost model. Our primary finding highlights the significance of SHAP in enhancing the interpretability and explainability of loan approval models, providing financial institutions with a valuable tool to ensure more transparent, reliable, and informed decision-making in the lending process.

Index Terms—Loan Approval, SHAP, Catboost, Machine learning

I. INTRODUCTION

In an increasingly data-driven financial landscape, the ability to make accurate and fair lending decisions is paramount. Traditional loan approval processes, often reliant on rigid criteria such as credit scores, income levels, and subjective assessments, are becoming less effective in capturing the complexity of borrowers' financial profiles. These methods can

introduce inefficiencies, errors, and biases, leading to unequal access to credit and hindering economic growth and financial inclusion.

With the advent of machine learning (ML), there is a growing opportunity to enhance loan approval systems by analyzing large, multidimensional datasets to uncover hidden patterns that traditional methods may overlook. ML can improve the accuracy, efficiency, and fairness of loan approval processes by capturing complex interdependencies between borrower demographics, credit history, and economic indicators.

However, the application of ML in financial decision-making raises critical concerns regarding transparency and interpretability. Many ML models, especially deep learning, function as "black boxes," providing predictions without clear explanations. This lack of transparency can hinder trust among stakeholders, especially lenders and borrowers, who need to understand how decisions are made. Without interpretability, stakeholders may not trust the system, leading to dissatisfaction and perceived unfairness. For financial institutions, ensuring interpretability is essential for regulatory compliance and fostering consumer trust.

This is where SHAP (SHapley Additive exPlanations) becomes crucial. SHAP enhances the transparency of ML models by explaining the contribution of each feature to the final prediction, helping stakeholders understand how specific attributes, such as income or credit score, influence loan approval outcomes. This interpretability is vital for fairness, accountability, and transparency, especially in high-stakes financial decisions.

In this study, we contribute by developing a transparent loan approval prediction model that integrates SHAP-based interpretability with advanced ML techniques. Our approach improves prediction accuracy by leveraging a comprehensive dataset, including borrower demographics, credit history, and

economic indicators, while ensuring the decision-making process is fully interpretable. We utilize SHAP to explain how individual features influence predictions, providing insights into the factors driving loan approval decisions. By combining ML with SHAP, our framework promotes fairness, accountability, and transparency in the loan approval process.

II. LITERATURE REVIEW

An overview of earlier research on Loan approval prediction is provided in this section. The basis for subsequent explorations in the field has been established by this study.

Ahmad S. et al. presented a prediction system for Loan Approval using ML Algorithm which utilizes a Kaggle dataset with 1500 cases, featuring 10 numerical and 8 categorical attributes related to bank customers. The primary technique employed is logistic regression, achieving a maximum accuracy of 81.1% on the test set. The study emphasizes the importance of customer attributes, such as credit score and income, in determining loan eligibility, indicating that higher incomes and lower loan amounts correlate with higher approval rates.

Uddin et al. applied multiple machine learning techniques, including Logistic Regression, Decision Trees, Random Forest, Extra Trees, and Support Vector Machines, along with deep learning models such as DNN, LSTM, and RNN to predict the loan approval. To address dataset imbalance, the SMOTE technique was employed. The ensemble model, combining the top three machine learning models, achieved an accuracy of 87.26%, outperforming individual models.

Viswanatha V. et al. utilized four machine learning algorithms to classify loan approval status including Random Forest, Decision Tree, Naive Bayes, and K-Nearest Neighbors (KNN). Among these, Naive Bayes achieved the highest accuracy at 83.73%, followed by Random Forest and KNN at 77.23%, and Decision Tree at 63.41%. The research contributes by providing a comparison of models and suggesting Naive Bayes as the most effective approach for loan approval prediction.

Teuku R. N. et al. examine how an interpretable machine learning approach using the LightGBM model can enhance the decision making process for loan approvals. The research employs a Random Search for hyperparameter tuning and the SHAP framework to improve model accuracy and interpretability. They compared the LightGBM model with four other ML models (Decision Tree, Random Forest, AdaBoost, and Extra Trees). The LightGBM model achieved an accuracy of 98.13%, outperforming other models.

Ugochukwu. E. et al. analyze loan eligibility by implementing six machine learning algorithms. The Random Forest model achieved highest accuracy at 95.55%, while Logistic Regression recorded the lowest at 80%. To address the common issue of imbalanced classes in credit risk assessment, they implement the Synthetic Minority Oversampling Technique (SMOTE), resulting in improved model performance.

III. METHODOLOGY

In this section, we briefly describe the methodology employed in this research, which includes data acquisition and preprocessing, CatBoost and other ML models, the integration of SHAP for interpretability, and the evaluation of performance metrics.

A. Data acquisition and Preprocessing

The dataset utilized in this research was collected from Kaggle[11], a very well-known platform for machine learning datasets. This dataset includes 36 features, with the target variable representing the binary classification of loan approval status, labeled as approved (0) or rejected (1). It contains 20,000 instances, of which 15,220 are classified as approved and 4,780 as rejected.

To prepare this dataset for model training, a careful preprocessing stage was conducted. This involved checking for any missing values and applying suitable methods, like imputation or removal to address them. Categorical features were transformed into a machine readable format through one-hot encoding and label encoding. The dataset was further divided into training and testing sets, with 80% (around 16,000 records) allocated for training the model and the remaining 20% (approximately 4,000 records) set aside for testing. This split allows the model to be trained on a large portion of the data while maintaining a separate dataset for an unbiased evaluation of its performance.



Fig. 1: Dataset distribution

B. CatBoost Classifier

In this study, CatBoost model was chosen as the primary model for its efficient handling of categorical features, fast training speed, and strong performance on imbalanced datasets, making it ideal for loan approval classification. Its compatibility with SHAP also facilitates transparent and interpretable insights essential for decision-making in finance. CatBoost (Categorical Boosting) is a gradient-boosting algorithm designed to handle categorical features efficiently and overcome common issues such as overfitting and target leakage. It uses ordered boosting, which orders the data points during the training of each decision tree to prevent target leakage. In ordered boosting, a new training dataset is obtained in each step of boosting, ensuring that the model obtained previously is applied to a set of new training samples. This

guarantees that the previous model has not seen the labels in the new training set, resulting in an unbiased model at each boosting step. To better illustrate the ordered boosting concept, assume M_t is our current model in the t -th iteration, which is learned using only the first t samples. The residual $r^{t+1} = y_i - M_t$ is unshifted because the model is trained without observations from the new training set. This process is supported by the generation of random permutations of training samples, ensuring that future data does not leak into the model during training, leading to better generalization. This concept is visually explained in Figure 1, where the model is trained incrementally on different subsets of data, avoiding any lookahead bias during training.

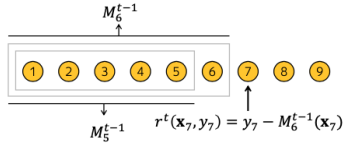


Fig. 2: Ordered boosting principle

Additionally, CatBoost handles categorical features by converting them into numerical values using statistics from subsets of the data, avoiding the need for one-hot encoding and preserving the relationships between categories. The algorithm builds an ensemble of decision trees, trained sequentially, with each tree correcting the residuals of the previous one. The model optimizes a loss function, typically in the form:

$$L(y, \hat{y}) = \sum_{i=1}^n \ell(y_i, \hat{y}_i) \quad (1)$$

where L is the total loss, y_i is the true label, and $\hat{y}_i$ is the predicted label for each instance i . For binary classification tasks, CatBoost often uses the Log-Loss function:

$$\ell(y_i, \hat{y}_i) = -(y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)) \quad (2)$$

CatBoost minimizes this objective by iteratively improving predictions. At each iteration, a new decision tree is added to the ensemble to reduce the residual errors of the previous iteration.

C. Shapley Additive Explanations (SHAP)

SHAP, introduced by Lundberg and Lee in 2017, is an explanation method based on cooperative game theory. It determines the individual contribution of each feature by calculating Shapley values, which represent the average marginal effect of a feature on predictions across all possible coalitions. Let f be the machine learning model to explain, g the explanatory model, and x and x' the input and simplified input, respectively. SHAP uses additive feature attribution to interpret the model's output as:

$$f(x) = g(x') = \phi_0 + \sum_{i=1}^M \phi_i x'_i \quad (3)$$

Here, M is the number of features, ϕ_i is the SHAP value for feature i and ϕ_0 represents the baseline value when all inputs are missing. SHAP provides both local and global explanations, where local explanations are combined to produce a global explanation by averaging the absolute Shapley values for each feature across the dataset. Additionally, SHAP reveals how each feature impacts predictions (positively or negatively) and measures interactions between features i and j .

D. Other ML Models

Several machine learning models were implemented alongside CatBoost to provide a comprehensive comparison. Gradient Boosting (GB) is an ensemble method that reduces bias and variance by sequentially building models, while LightGBM (LGBM) is optimized for speed and efficiency with large datasets. The Bagging Classifier improves accuracy by averaging predictions from multiple base learners, typically decision trees, and AdaBoost (AB) focuses on adjusting weights of misclassified instances to improve weak learners. XGBoost offers optimized gradient boosting with regularization to prevent overfitting, and Random Forest (RF) builds an ensemble of decision trees on random subsets for greater predictive accuracy. Extra Trees (ET) introduces randomized splits for computational efficiency, while Decision Tree (DT) provides an easy-to-interpret model based on recursive splits. These models serve as benchmarks for evaluating the performance of CatBoost in predicting loan approvals.

E. Evaluation Indexes

The performance of the Catboost model was compared against eight ML models, and the models were evaluated using two key metrics: the coefficient of determination (R^2) and the root mean square error (RMSE). RMSE clearly indicates the residual error for each sample, giving us a precise view of how well the models are performing. On the other hand, R^2 normalizes the squared residual error with the variance in the dataset, resulting in a score between 0 and 1.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (4)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}} \quad (5)$$

This makes R^2 an intuitive choice for comparing different models. In the end, R^2 was chosen as the main metric for further analysis.

IV. RESULTS AND DISCUSSION

A. Model Performance

To assess the generalization capability of the nine machine learning models, each model was fine-tuned using optimal hyperparameters. The determined hyperparameters were assigned to the machine learning models, and the training dataset was utilized to train the models. Subsequently, model testing was conducted on the testing dataset. The performance metrics of the classifiers, including R^2 , and Root Mean Square Error (RMSE), are summarized in Table I.

TABLE I: Comparison of performance metrics of the classifiers.

Models	R^2		RMSE	
	Train	Test	Train	Test
CatBoost	0.9976	0.9921	0.0209	0.0387
XGBoost	0.9927	0.967	0.0362	0.0791
LGBM	0.9684	0.9552	0.0754	0.0922
Random Forest	0.9965	0.9486	0.025	0.0987
Gradient Boosting	0.8714	0.8892	0.1521	0.1449
Extra Trees	1.0	0.9037	0.0	0.1351
AdaBoost	1.0	0.9459	0.0	0.1012
Decision Tree	1.0	0.9473	0.0	0.1
Bagging	0.9983	0.9684	0.0177	0.0775

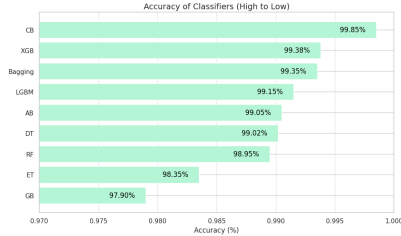


Fig. 3: Accuracy of Classifiers

The impressive R^2 value of the CatBoost model indicates a strong correlation between predicted and actual values, reinforcing its role as a reliable predictor in this analysis. The RMSE values further validated the models' predictive performance, demonstrating minimal deviation from actual outcomes.

The accuracy of the models was notably high, with the CatBoost Classifier achieving the highest accuracy of 99.85%. In comparison, the other classifiers also performed well, with accuracies as follows: XGBoost at 99.38%, Bagging Classifier at 99.25%, and LightGBM at 99.15%. Other classifiers such as AdaBoost (98.98%), Random Forest (98.95%), Decision Tree (98.92%), Extra Trees (98.22%), and Gradient Boosting (97.90%) also showed strong performance metrics. Figure 3 illustrates the comparative accuracy of all classifiers in a bar chart, clearly highlighting the CatBoost model's superior performance. The corresponding precision, recall, and F1 score further underscore CatBoost Model's effectiveness in predicting loan approvals, as depicted in Figure 4.

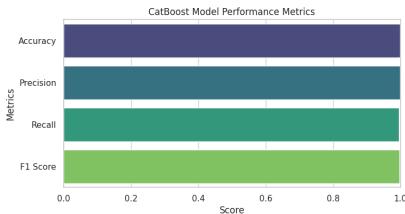


Fig. 4: Catboost model performance matrices

Additionally, the CatBoost model's confusion matrix is displayed in Figure 5, which provides insights into the model's classification performance, revealing the number of true positives, true negatives, false positives, and false negatives.

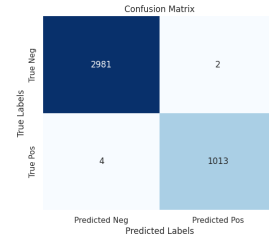


Fig. 5: Confusion Matrix of CatBoost Model

To visualize the comparative performance of the classifiers, radar charts were drawn to represent the evaluation metrics across classifiers, as shown in Figure 6. This visualization highlights the strengths and weaknesses of each classifier across different performance metrics.

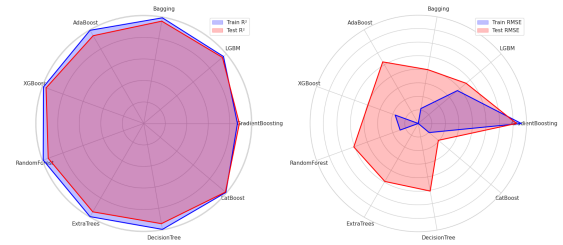


Fig. 6: Radar chart for R^2 and RMSE

Lastly, a Taylor diagram of the testing dataset was plotted to compare the statistical relationships between the predictions of the various models and the experimental data, incorporating R^2 and RMSE. This is illustrated in Figure 7, emphasizing the models' predictive capabilities and their alignment with actual outcomes.

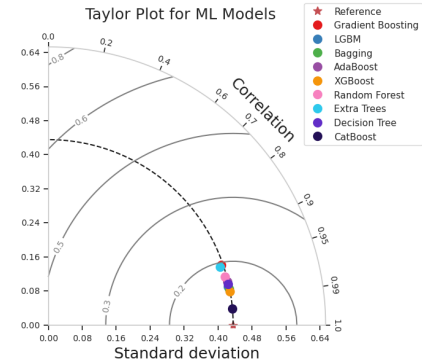


Fig. 7: Taylor diagram of the testing dataset.

B. Model Interpretability

SHAP (SHapley Additive exPlanations) was utilized to interpret the CatBoost model's predictions, offering both global and local insights into the model's decision-making process.

1) *Global Interpretability*: To evaluate how each feature contributes to the overall predictions, SHAP global feature importance was analyzed. The global feature importance plot

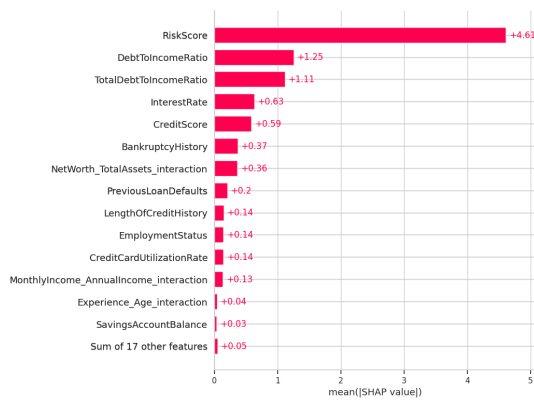


Fig. 8: Feature Importance Plot using Shapely Values

(Figure 8) shows the average impact of each feature on the model's output, measured using SHAP values. As observed, the RiskScore was the most influential feature, followed by TotalDebtToIncomeRatio and DebtToIncomeRatio, indicating that financial stability metrics played a key role in predicting loan approvals. Other features, such as InterestRate and CreditScore, also contributed significantly to the model's performance, though to a lesser extent.

The SHAP beeswarm summary plot (Figure 9) illustrates the global impact of each feature on the CatBoost model's predictions. It displays the distribution of SHAP values for all features, with colors representing feature values (red for high, blue for low). In this plot, the RiskScore once again stands out as the most impactful feature, with a wide range of SHAP values showing its strong influence on the predictions. Notably, higher RiskScore values (in red) tend to increase the likelihood of loan approval. Similarly, TotalDebtToIncomeRatio and DebtToIncomeRatio demonstrate significant impacts on the model's predictions, with both high and low values influencing the model's decisions in different directions.

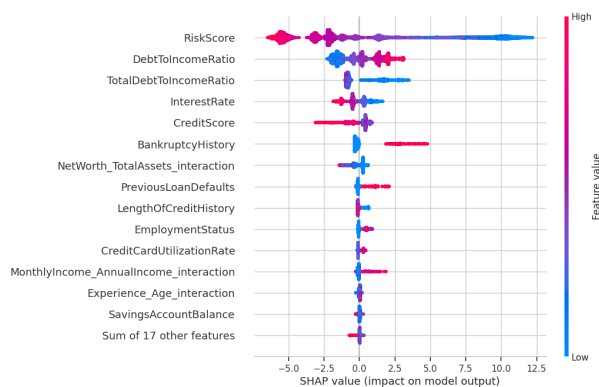


Fig. 9: Beeswarm summary plot using Shapely Values

The SHAP dependence plot (Figure 10) illustrates the interaction between RiskScore and the MonthlyIncome AnnualIncome interaction, highlighting how changes in RiskScore impact its SHAP value and the model's predictions. The plot reveals that lower RiskScore values (around 30-40) are

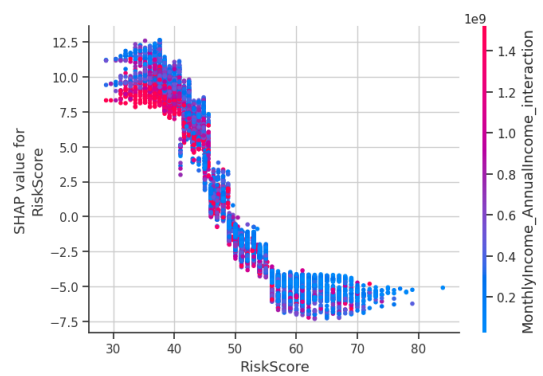


Fig. 10: Dependence Plot using Shapely Values

associated with positive SHAP values, indicating a higher likelihood of loan approval, while higher RiskScore values (above 60) correspond to negative SHAP values, suggesting a lower probability of approval. This non-linear relationship underscores the significance of RiskScore in the prediction process. Additionally, the color gradient from blue to red reflects how the MonthlyIncome AnnualIncome interaction influences the overall effect of RiskScore.

2) *Local Interpretability*: While global interpretability provides an overview, understanding individual predictions requires local analysis. SHAP enables this by offering insights into how each feature contributes to a specific prediction.

The SHAP waterfall plot (Figure 11) offers a detailed view of how the model's final prediction for a specific instance is determined, beginning from a baseline value (the average prediction) and illustrating the impact of each feature. In this plot, the RiskScore feature significantly decreases the predicted output by -5.95, indicating that a lower risk score correlates with a lower final prediction, while other features such as DebtToIncomeRatio and InterestRate, also negatively influenced the prediction, pushing it further downward (towards approval). Conversely, features such as BankruptcyHistory and the NetWorth TotalAssets interaction contribute positively, with BankruptcyHistory adding +2.79 to the final prediction, suggesting that a history of bankruptcy is associated with a higher likelihood of negative credit outcomes (towards rejection).

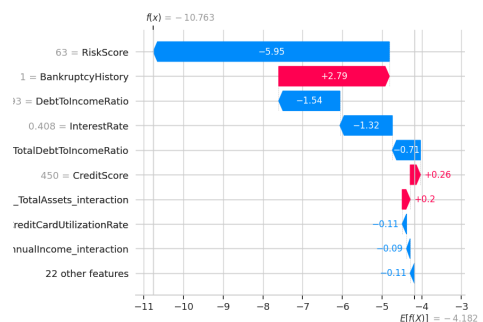


Fig. 11: Waterfall Plot using Shapely Values



Fig. 12: Force Plot using Shapely Values

The SHAP force plot (Fig. 12) provides a compact visualization of how specific features influence the prediction for an instance, functioning similarly to a waterfall plot but in a more condensed format. The base value, representing the average model prediction, is -4.182, while the final predicted output is -6.87, indicating that the combined effect of the features led to a prediction lower than the baseline. On the right side of the plot, features like RiskScore and InterestRate contributed negatively to the prediction, with RiskScore = 58 having the most substantial downward effect. Conversely, the left side of the plot highlights positive contributions from features such as the NetWorth TotalAssets interaction and CreditScore, with the NetWorth and TotalAssets interaction being the most influential positive factor, significantly boosting the final prediction.

The SHAP decision plot (Figure 13) provides a high-level overview of the cumulative effects of features on the model's prediction for a specific instance, displaying SHAP values on the X-axis. Key features like RiskScore, DebtToIncomeRatio, and TotalDebtToIncomeRatio significantly impact the prediction, with RiskScore having a strong negative effect, while BankruptcyHistory contributes positively. The plot also highlights interaction terms, such as NetWorth TotalAssets and MonthlyIncome AnnualIncome, demonstrating the model's ability to capture complex relationships between features.

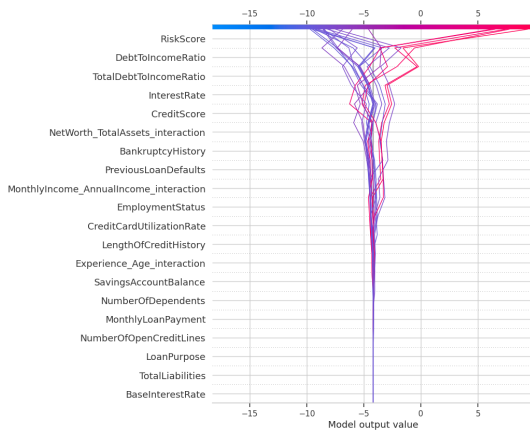


Fig. 13: Decision Plot using Shapely Values

V. CONCLUSION AND FUTURE WORK

In this study, we employed CatBoost as the main machine learning model to predict loan approvals, combining it with SHAP (SHapley Additive exPlanations) for model interpretability. CatBoost demonstrated superior performance, with SHAP identifying critical features such as RiskScore, DebtToIncomeRatio, and TotalDebtToIncomeRatio as key drivers

of loan outcomes. By using SHAP, we could offer both global and local interpretability, revealing how individual factors contributed to each decision. This approach fosters transparency in financial decision-making, ensuring that stakeholders, including lenders and borrowers, can trust the system's fairness and understand the rationale behind loan approvals or rejections. Our work aims to address the increasing demand for fairness and transparency in the financial sector, particularly in loan approval processes. By identifying and explaining key factors influencing outcomes, the model supports more equitable lending practices, helping mitigate bias and build trust. Additionally, the adaptability of the CatBoost model ensures that it can remain relevant as economic conditions evolve, providing ongoing value to financial institutions. This study contributes to the growing body of research on interpretable machine learning, emphasizing the critical role of explainability in sectors with significant socio-economic impact, such as finance. Future work could extend this research by exploring other interpretability frameworks like LIME or ALE, further promoting fairness and accountability in financial decision-making.

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